Automatic Detection of Asphalt Pavement Crack Width Based on Machine Vision

Rui Tao, Rui Peng, Yong Jin, Fangyuan Gong, and Bo Li

***Abstract*— Pavement disease is a key factor affecting road safety, and regular inspection and maintenance play an important role in maintaining the performance of asphalt pavement. Manual crack detection is subjective and inefficient. Even with the rapid development of pavement detection technology in recent years, it is still a challenging task to detect asphalt pavement cracks from images due to the complex environment. Crack classification and identification based on deep learning have achieved many results, but there are still deficiencies in terms of parameters and model efficiency. Aiming at the above problems, this paper proposes a new crack identification and analysis system guided by the structured information in the intelligent asphalt pavement disease image. First, the pavement crack types are classified by applying the convolutional neural network (CNN) algorithm to mine images containing 4,000 pavement breakage images and using the sparrow search algorithm (SSA) to improve the accuracy of the model. Aiming at the common problems of low contrast, serious noise pollution, and uneven illumination in the special environment of the tunnel, the improved Retinex enhancement algorithm is used to enrich the image details. Affected by the different textures of pavement materials, imaging conditions, etc., it is proposed for the first time to calculate the crack width through the skeleton boundary refinement algorithm. The results of experimental tests and case studies show that the proposed approach demonstrates the potential of innovative deep learning methods in automatically calculating pavement quality indices to optimize the repair and maintenance of asphalt pavements.**

***Index Terms*— Asphalt pavement, crack classification, SSA-CNN, image processing, road maintenance.**

I. Introduction

WHILE the rapid development of highways has brought convenience to our daily lives, a new issue has emerged: highway maintenance [1]. The degradation of pavement problems such as cracks, rutting and potholes is accelerated by numerous factors in daily life, such as heavy loads and temperature changes. Regular pavement condition assessment and maintenance is essential to ensure the functionality of the pavement [2]. Cracks are one of the most common and harmful pavement diseases, which will reduce the bearing capacity and water resistance of the pavement, thereby accelerating the deterioration of the pavement. If the cracks are not found and repaired in time, it will affect the performance and normal service life of the road, and reduce the comfort and safety of the road [3]. In addition, if cracks are not treated for a long time, they will become potholes, which may deform the road surface, and even lead to varying degrees of pavement peeling [4]. Crack repair methods and costs vary greatly depending on the severity of the crack, and proper repair and maintenance can prevent serious damage from cracks. Therefore, timely and precise pavement crack identification is crucial for maintaining the pavement and preventing accidents.

Pavement cracking is a common pavement distress that can lead to path water infiltration, road base instability, and even serious damages such as road base collapse [5]. Therefore, it is essential to identify and classify pavement cracks. According to Highway Performance Assessment Standards (JTG5210-2018) in China, cracks are categorized into transverse cracks, longitudinal cracks, alligator cracks and block cracks [6]. Transverse cracks refer to cracks that are basically perpendicular to the direction of driving, mainly caused by temperature changes causing expansion and contraction, as well as small movements and deformations caused by traffic loads. Longitudinal cracks refer to cracks that are basically parallel to the direction of travel. When paving the asphalt surface layer, two residues were not properly treated, and longitudinal cracks gradually formed under the influence of vehicle loads and environmental factors. Due to uneven compaction or water immersion of the roadbed, uneven settlement at the edge of the roadbed can also lead to the formation of longitudinal cracks [7]. In addition, vehicle loads, environmental climate, and other factors can cause various degrees of cracking in the pavement surface and base. However, simple crack identification and morphology feature extraction have no practical guidance for maintenance decisions.

Cracks in road surface images can be defined as a set of pixels with lower intensity than the surrounding background, forming a curve with different widths and directions [8]. Rule based methods are methods that use predefined features to identify or represent cracks in an image, based on an important assumption that crack pixels are always continuous and darker than the surrounding environment. On the morphological side, structural features can be used for detection. Ouma et al. used discrete wavelet transform and progressive morphological

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transform filtering to extract linear damage shape and continuity [9]. Oliveira et al. proposed a method to estimate the average crack width by calculating the ratio of the crack area and the number of crack skeleton pixels, but the method is only applicable to the case of uniform pavement crack width, which usually exhibits heterogeneity in most practical situations [10]. Liu et al. developed a method for calculate the crack width by extracting the centerline and edge of the crack, but this method cannot truly reflect the severity of cracks in the local neighborhood [11]. Qiu et al. improved the calculation method of the crack width in the pixel direction, and determined the measurement direction of each crack pixel by using orthogonal projection, thereby improving the measurement accuracy [12]. However, orthographic projections can produce intersection artifacts when determining the measurement direction, resulting in blurred measurements at the intersection locations. Similarly, Wang et al. proposed a new method using the electric field model of the Laplace equation to calculate the direction of width measurement for each crack pixel, but these crack width measurements did not take into account the severity of the crack [13]. Kang et al. used a modified distance transform method to measure crack thickness and length, using various bounding boxes and a modified tubular flow field (TuFF) algorithm to segment crack pixels from the detected crack region used for localization [14], but it is susceptible to light, shadows, and various noises.

In order to overcome the limitations of traditional image processing-based damage detection methods, deep learning based methods were introduced. Cha et al. pioneered the detection of concrete surface cracks by designing a deep convolutional neural network (CNN) [15]. Based on this, an ultrasonic beacon system was utilized to geo-tag UAVs for concrete crack detection [16]. However, this technique requires significant computational cost and needs to consider the characteristics of the camera and lens, the distance between the camera and the object, and the size of the crack to determine the appropriate sliding window size. Instead of the sliding window method, a faster region-based convolutional neural network (FAST-R-CNN) is used for damage detection and localization, and various sizes of bounding boxes are proposed to detect and localize different sizes of damage [17]. The is faster than other types of localization methods (e.g., sliding window techniques) and has become the mainstay of the deep learning-based multiple types of damage detection problem. However, it is susceptible to light, shadow and various noises. Several semantic segmentation algorithms have achieved significant results [18], [19], [20], [21], such as pixel-level segmentation methods to quantify the detected damage. This real-time crack segmentation method has an average mIoU of 0.846 with 160,000 parameters and 37 frames per second (FPS) [22]. However, the computational cost of the algorithm parameters is still high, and most of the pavement distress detection is a field-collected dataset, which requires neural networks with fewer parameters to reduce the computational cost [23]. Attention-based Generative Adversarial Network (AGAN) to generate internal damage images of synthetic concrete elements developed IDSNet outperforms other state-of-the-art networks and is capable of real-time processing of thermal images of 640 × 480 × 3 size at 74 FPS [24], but generalizability to different types of defects has not been investigated. Rapid processing is an important aspect of pavement disease monitoring, as the length of highway mileage is large and requires processing a large number of images to detect pavement diseases and conduct comprehensive pavement grading. It may not necessarily be processed in real-time, but it needs to reduce overall computational costs and provide rapid updates of road surface status.

Zhu et al. proposed an efficient lightweight encoder-decoder network embedded in a developed mobile robot system that can effectively detect pavement cracks in real scenes at 25 frames per second [25].Kang and Cha proposed an algorithm for assessing the complexity of complex scene images with structural damage, where the basic concept is to count the number of objects in the scene [26]. The larger the number of objects, the higher the complexity level. However, image complexity is a multidimensional concept, and it is usually necessary to consider multiple methods and factors to comprehensively evaluate image complexity. The specific evaluation methods may vary depending on the research purpose and field. In road engineering, since the pavement distress data acquisition in real conditions is affected by various light sources and weather, plus the different acquisition equipment will also affect the shooting results, it is inevitable that there will be uneven lighting and unclear local details in the images obtained, such as pavement markings, rutting, shadows, scratches, and rough backgrounds [27], [28]. In particular, the complexity of the pavement background in tunnels leads to intensity inhomogeneities of the cracks, e.g., low contrast with the surrounding pavement and shadows of similar intensity, which remains a challenging task. Recent research has developed crack segmentation methods that implement deep learning, and although most of the published methods achieve desirable detection results [5], [29], [30], they only segment cracks in a monotonous background. The image datasets collected from tunnel pavements usually include non-crack images disturbed by light, noise, etc., and these methods may lead to false detection. The above deep learning algorithms only focus on the crack image and only focus on what the crack object is, while ignoring the basic spatial structure and positional information in the crack image, resulting in incorrect edge crack localization and poor anti-interference ability. For road engineering, the calculation of crack morphological characteristics should be transformed into relevant indicators to guide the maintenance decision and construction of roads.

In order to solve the above challenging problems, the main contributions of this work can be summarized as follows: (1) Optimize the hyperparameters of the CNN using the SSA to reduce the computational cost of parameter selection, classify the types of pavement cracks according to the constructed SSA-CNN model, and test its detection effect. (2) An image enhancement algorithm is proposed for the obvious differences between road tunnel crack images and road crack images. This image enhancement algorithm is used to highlight the characteristics of the object information, minimize the noise, improve the clarity of the details, and make the image more conducive

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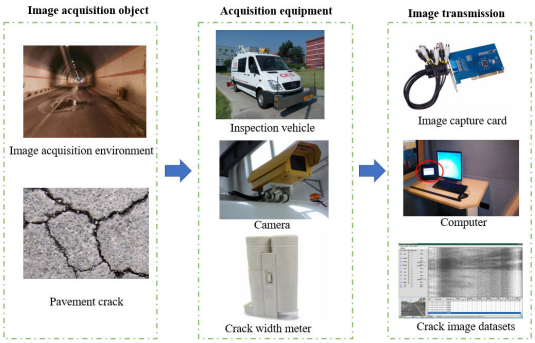

Fig. 1. Data collection and processing.

to subsequent analysis and application. (3) Affected by the different textures of pavement materials, imaging conditions, etc., the skeleton boundary refinement algorithm is proposed for the first time to avoid the fluctuation of the edge noise of crack images on the measurement of pixelated crack width.

The rest of this paper is structured as follows. Section II describes the data acquisition. Section III is the methodology. Section IV is the area calculation of the crack morphology, and Section V verifies the effectiveness of the proposed method through experiments. Finally, the results achieved in this paper are summarized, and the future work is prospected.

II. Image Acquisition Procedure and Dataset Description

Shanxi Province is located in the north of China, with a mountainous plateau landscape, mostly loess geology and low winter temperatures. As a major coal-producing province, Shanxi Province has a large proportion of heavy vehicles and heavy daily traffic, resulting in serious crack disease in most sections and deteriorating pavement quality. The object of this research is the two highways of Taichang and Qilin. A multifunctional road detection vehicle is used as the data collection platform, and a SONY-SELP18105G high-definition camera is installed as the data sensor. The scan was performed to provide full coverage of the 3.75 m wide pavement, taking an image of the crack every 15 s. The collected images are converted from analog signals to digital signals by the A/D module, and finally special image processing is carried out. The processed digital signals are saved in an external memory, and these image data can be used for subsequent fracture dataset creation and analysis. As shown in Fig. 1.

After performing the above operations, 2000 pavement images with a resolution of 512 × 512 pixels were obtained. Training large images may lead to excessive GPU memory usage and result in training failure. To avoid this, and to achieve a good compromise between the computational effort and the accuracy of the test results [31], the large images are cropped to 256 × 256, facilitating the training using multiple batches of simultaneous network inputs. The trained network can test any image larger than the training image block. However, using an image smaller than this would make it more difficult to mark as defective or complete, so the original image was cut into 3-channel (RGB) 256 × 256 pixel blocks to create the crack dataset.To increase the diversity of the training data, the training images were rotated, panned, clipped, and scaled with a rotation angle between 0-20 degrees, a translation range of 0-5% of the image length and width, a clipping intensity of 0-0.2 radians, and a scaling range of ±5%. These transformation parameters are randomly selected to perform a full range of data enhancement operations within the specified range of values, and a real-time data enhancement training sample generator is constructed, thus greatly expanding the number of training samples.

TABLE I
Asphalt Crack Image Datasets

Task	Classification	Enhancement	Morphometry
No. of images	4000	1200	800
Size (pixels)	256×256	256×256	256×256
Training set	2400	-	-
Validation set	800	-	-
Test set	800	-	-

During the training process, in order to avoid the insufficient number of crack images that can easily lead to the overfitting problem. Therefore, data enhancement is realized by increasing the collected 2000 data images to 4000, which are divided according to 60% of training set, 20% of validation set and 20% of testing set; 1200 images are collected in the internal environment of the tunnel for image enhancement experiments; in order to test the measurement accuracy of the crack algorithm proposed in this paper, the width of 800 crack images is measured by using a crack width meter, as shown in Table I.

III. Method

This study is divided into three parts. First, the test vehicle is equipped with a camera to collect data and build the database. The CNN model including convolution layer, pooling layer, fully connected layer, and softmax layer is optimized through sparrow search algorithm to achieve crack classification and improve model performance. Secondly, according to the special characteristics of the environment inside the tunnel, the tunnel pavement image is uniformly enhanced by using the improved multi-scale Retinex. Finally, the image morphology parameter extraction model is established, and the width of the linear crack and the width of the network crack are extracted and the area is calculated by the skeleton refinement algorithm. The damage degree of the crack can be further judged by grading the damage degree of the crack according to the current specification. The technology roadmap is shown in Fig. 2.

A. CNN

CNN is widely used in the field of image recognition as a kind of pre-feedback neural network [32]. It uses iterative algorithms such as forward propagation and backward error propagation to reduce the error between the predicted value and the actual value. During the forward propagation process, CNN will initialize the weights and biases, and perform

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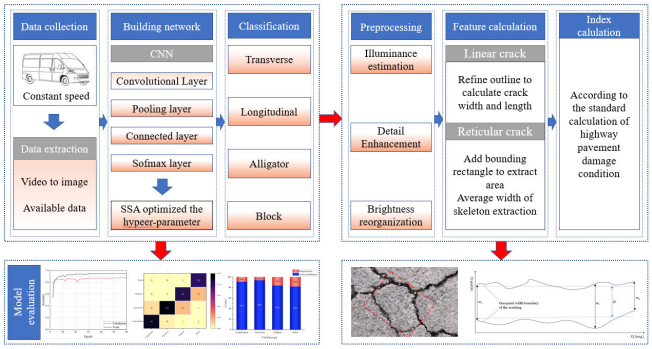

Fig. 2. Workflow of automatic crack detection.

dimensionality reduction operations such as convolution and pooling on the input image data to extract features in the image. Through this process, CNN can efficiently complete various image classification and processing tasks. In order to test the performance of the crack image classification model, the model needs to be evaluated using relevant metrics. The input layer is the first layer of the network, and the image is transformed into a matrix as the input to the next convolutional layer, which is a crack image of size 256 × 256 pixels. Conv1 to Conv3 use 3 × 3 convolutional kernels, with Conv1 having 32 kernels and Conv2 and Conv3 doubling the number of kernels one layer at a time to 64 and 128 respectively, and each layer is followed by a ReLU activation function. The three pooling layers (poo1, pool2 and pool3) use 2 × 2 kernels with a step size of 1. The CNN structure designed in this paper has two fully-connected layers, which connect the high-level features learned in the convolutional layers and are placed after the third pooling layer and before the Softmax layer. In the final layer of the CNN, the crack images and their feature vectors are classified by a Softmax classifier.

B. SSA

The sparrow search algorithm is a new type of group intelligence optimization algorithm, whose main idea is derived from the foraging and anti-predatory behavior of sparrow groups [33]. Since sparrows are a flock species with strong memory and vigilance, there is a clear division of labor in the sparrow population when foraging, which is roughly divided into three roles: discoverer, joiner and scout. The discoverer is responsible for searching for food and guiding the entire sparrow population in the area and direction of foraging, the follower follows the discoverer to forage, and the scout sparrow is responsible for sensing danger and will sound an alarm in time to prompt the sparrow population to shift its position.

1) Discoverer Location:

$$X_{i,j}^{t+1} = \begin{cases} X_{i,j}^{t}\exp(-\dfrac{i}{\alpha iter_{max}}), & R_2 < ST \\ X_{i,j}^{t} + Q\mathbf{L}, & R_2 \geq ST \end{cases} \tag{1}$$

where t is the number of current iterations, $j = 1, 2, 3\ldots n$, $iter_{max}$ is a constant indicating the maximum number of iterations, $X_{i,j}$ is the position information of the i-th sparrow in the j-th dimension, $\alpha \in (0, 1)$ is a random number, $R_2 \in (0, 1)$ and $ST \in [0.5, 1]$ are the warning and safety values, respectively, Q is a random number obeying normal distribution, and $\mathbf{L}$ is a $1 \times d$ dimensional matrix, d represents the dimension.

2) Joiner Location:

$$X_{i,j}^{t+1} = \begin{cases} Q\exp(-\dfrac{X_{\text{worst}} - X_{i,j}^{t}}{i^2}), & i > \dfrac{n}{2} \\ X_p^{t+1} + \mid X_{i,j}^{t} - X_p^{t+1} \mid \mathbf{A}^{+}\mathbf{L}, & i \leq \dfrac{n}{2} \end{cases} \tag{2}$$

where X_p is the optimal position occupied by the current discoverer, X_{worst} is the current global worst position, and $\mathbf{A}$ is a $1 \times d$-dimensional matrix, with each element randomly assigned a value of 1 or - 1, and $\mathbf{A}^{+} = \mathbf{A}^{T}(\mathbf{A}\mathbf{A}^{T})^{-1}$. When $i > \frac{n}{2}$, it indicates that the i-th participant with a low fitness value did not receive food and is in a very hungry state, requiring them to seek food elsewhere.

3) Scout:

$$X_{i,j}^{t+1} = \begin{cases} X_{\text{best}}^{t} + \beta \mid X_{i,j}^{t} - X_{\text{best}}^{t} \mid, & f_i > f_g \\ X_{i,j}^{t} + K(\dfrac{\left|X_{i,j}^{t} - X_{\text{worst}}^{t}\right|}{(f_i - f_w) + \varepsilon}), & f_i = f_g \end{cases} \tag{3}$$

where X_{best}^{t} is the current global optimal position, β is the step control parameter, obeys a normally distributed random number with mean 0 and variance 1, $K \in [-1, 1]$ is a random number, f_i is the current fitness value of the individual sparrow, and f_g and f_w is the current global optimal and worst fitness values, respectively, and ε is a constant to avoid 0 in the denominator.

C. SSA-CNN Model

Existing networks require a large amount of unnecessary computational cost due to their excessive number of learnable parameters. The appropriate choice of hyperparameters directly affects the performance of the network and has a direct impact on the quality of the model. Configuring and selecting appropriate hyperparameters (e.g., learning rate and regularization parameters) is tedious, and there are no definitive guidelines available for the optimization of these parameters. In [15] and [17], these two optimal network architectures for concrete crack detection must be explored through trial and error, a time-consuming process and difficult to optimize. To address this problem, combining the SSA algorithm with CNN can improve network performance by optimizing the weights. Fig. 4 shows the flow of the SSA to optimize the CNN.

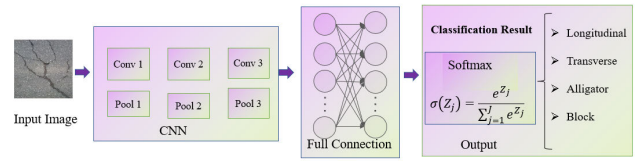

Fig. 3. CNN structure diagram.

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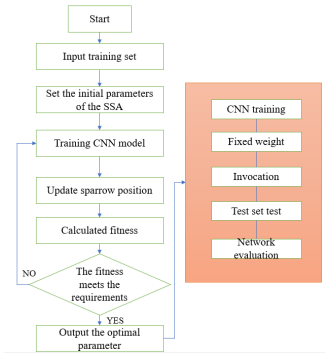

Fig. 4. SSA optimized CNN flow chart.

D. Multi-Scale Retinex Algorithm

Due to the special characteristics of the tunnel environment, road images are often affected by low visibility when images are taken in low light conditions. This low visibility not only affects the visual effect but also causes difficulties for subsequent use [34]. In order to solve the problems of poor image visibility and color deviation in low illumination conditions, this paper proposes an improved Retinex network enhancement method, which is based on the Retinex theory, to obtain the reflective image of the luminance channel. Then, using the different illumination results corresponding to the constraints at different scales of the luminance channel, the coarse details, medium details and fine details of the image are extracted respectively, and the reflection image is enhanced in detail by multi-scale detail weighting.

1) Illumination Estimation: The multi-scale Retinex algorithm calculates multiple scale factors for a single low-illumination image based on the single-scale Retinex algorithm, and linearly superimposes the results, as in Eq. (4):

$$R_i(x, y) = \sum_{i=1}^{k} W_k\{lg I_i(x, y) - \lg\left[F_k(x, y) * I_i(x, y)\right]\} \tag{4}$$

where W_k is the weight coefficient of different scales, and the sum of all scales W_k is 1, k is the number of Gaussian surround functions, generally three, representing high, medium and low scales, $F_k(x, y)$ is the scaling factor equal to the Gaussian surround function. Multi-scale Retinex algorithm can effectively maintain the detail features of the enhanced low-intensity image. $I_i(x, y)$ represents the i-th scale of the input low-light image, where (x, y) denotes the spatial location coordinates of the image.

The conventional multi-scale Retinex algorithm has a Gaussian surround function, which is prone to haloing after image processing. Compared with the Gaussian filter, the bilateral filter is a weighted non-linear filter, which can enhance image smoothing, preserve image edge information and suppress the appearance of halos while achieving a better denoising effect. Therefore, this paper uses the bilateral filter function instead of the Gaussian function as the surround function of the Retinex algorithm to improve the traditional algorithm. The definition of the bilateral filter function is:

$$l(x, y) = \frac{\sum_{m,n\in w_k} d(x, y, m, n)\,\lambda(x, y, m, n)\,f(x, y, m, n)}{\sum_{m,n\in w_k} d(x, y, m, n)\,\lambda(x, y, m, n)} \tag{5}$$

$$d(x, y, m, n) = e^{-\frac{(x-m)^2+(y-n)^2}{2\sigma_d^2}} \tag{6}$$

$$\lambda(x, y, m, n) = e^{-\frac{|f(x,y)-f(m,n)|^2}{2\sigma_r^2}} \tag{7}$$

where w_k is the bilateral filter window width; $f(x, y, m, n)$ is the pixel value centred on (x, y) in the domain; $f(m,n)$ represents the pixel value located at the pixel coordinates (m,n) in the image. σ_d and σ_r are the definition domain and value domain variance respectively. According to Eq. (7), the final luminance channel illuminance image can be obtained; m and n represent the horizontal/vertical pixel coordinates of the image, respectively.

2) Detail Enhancement: According to Retinex theory, the reflected image corresponding to the brightness channel can be obtained, which is recorded as V_r. The brightness of the reflected image has been improved to some extent. In order to make the details more prominent, high-frequency components can be applied to the reflected image to strengthen the ability of detail expression. Record the illumination images corresponding to different control factors as $V_l^{(1)}$ (illuminance 1), $V_l^{(2)}$ (illuminance 2) and $V_l^{(3)}$ (illuminance 3), multi-scale detail extraction is performed, and the expression is as follows:

$$\begin{cases} D_1 = V - V_l^{(1)}, \\ D_2 = V_l^{(1)} - V_l^{(2)}, \\ D_3 = V_l^{(2)} - V_l^{(3)} \end{cases} \tag{8}$$

$$D = \left[1 - \omega_1 \cdot sgn(D_1)\right] \cdot D_1 + \omega_2 \cdot D_2 + \omega_3 \cdot D_3 \tag{9}$$

In Eqs. (8)-(9), D_1 is coarse detail; D_2 is medium detail; D_3 is fine detail; D is the weighted detail of D_1, D_2 and D_3; ω_1, ω_2 and ω_3 are the weighting coefficients, which are 0.5, 0.5 and 0.25 respectively according to the literature [35]. V represents the original image.

After multi-scale detail extraction, the final weighted detail is superimposed with the reflected image of the brightness channel to enhance the detail of the reflected image. The detail enhancement result is recorded as $\tilde{V}_r$, the expression is as follows:

$$\tilde{V}_r = V_r + D \tag{10}$$

3) Brightness Reorganization: In order to improve the visibility of the output image, by normalizing the illuminance image, in this paper, the incident component $L(x, y)$ after bilateral filtering is re-processed by power law transformation correction to adjust the global brightness of the image. The expression is as follows:

$$L'(x, y) = s[L'(x, y)]^{\frac{1}{\gamma}} \tag{11}$$

where s and γ are correction constants. The parameter γ of the power-law transform controls the shape of the curve of the luminance adjustment and usually takes a value between

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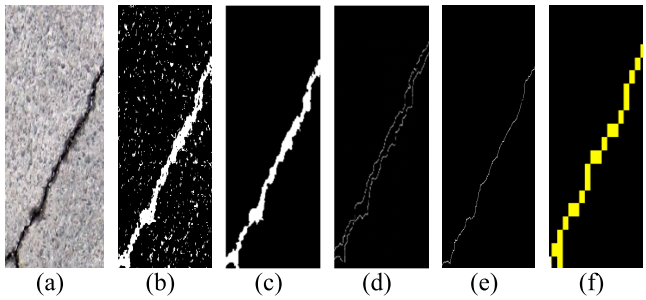

Fig. 5. Crack width measurement: (a) Original image; (b) Binarization; (c) Denoising; (d) Edge detection; (e) Skeletonization; (f) Morphological calculation.

$(0, +\infty)$. In applications, the value of s is usually selected based on the brightness distribution and needs of the image. If the image needs an overall increase in brightness, s can take a large positive value; if the brightness needs to be reduced, s can take a value less than 1, or even close to 0. Different application scenarios may require different values of γ and s to achieve optimal visualization.

IV. Crack Morphology Feature Extraction

One of the distinctive indications of the structural durability of asphalt pavements is crack width [36]. Monitoring the width of cracks in asphalt pavements allows engineers to determine whether immediate repairs are required and to develop a plan. To extract such information, the pixel predictions are post-processed in the following manner: crack images are post-processed using digital image processing methods to extract information on crack morphology and width, with specific steps including image binarization, connected domain denoising, edge detection, crack skeleton, and crack width measurement, as shown in Fig. 5.

A. Morphological Measurement

1) Image Binarization: As shown in Fig. 5(b), the maximum between-class variance method is used to convert the gray value of each pixel of the gray image to 0 or 255, reduce the amount of image data and highlight the target contour.

2) Connected Domain Denoising: Pavement crack images have a distinct distribution of two-pixel values, where one represents the crack and the other represents the background. The bimodal histogram usually indicates that there are two or more different sets of pixel values in the image, which is reasonable for both cracks and background. As shown in Fig. 5(c), the size of the number of pixels in each connected domain in the image needs to be traversed first, and then the connected domains with the number of pixels less than the threshold M are removed. The bimodal histogram analysis method can determine the range of pixel values for the cracks and the background, and the pixel values located between the peaks can be selected as the threshold value, and finally the broken connected domains will be connected.

3) Edge Detection: As shown in Fig. 5 (d), the laplacian edge detection method is used to find the edge of the crack, and the position of the edge is determined by finding the gradient direction of the gray change. Non-maximum suppression technology is used to eliminate edge misdetection, and double threshold method is used to find potential boundary, and lag technology is used to track the boundary.

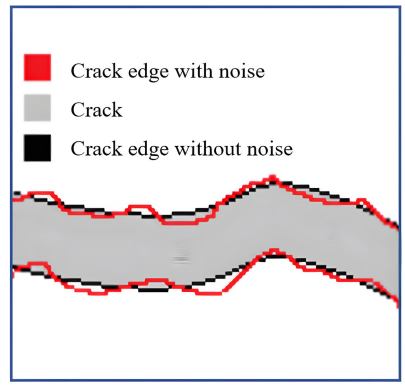

Fig. 6. Crack width measurement error due to noisy crack edges.

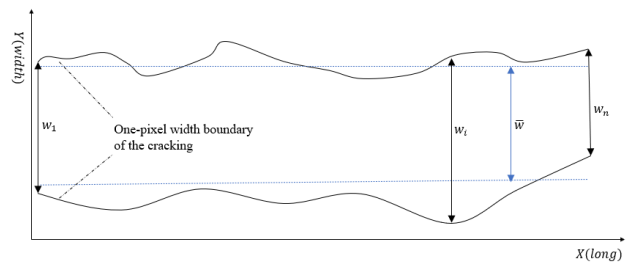

Fig. 7. Diameter statistics of extracted skeleton boundaries.

4) Skeletonization of Morphological Cracks: As shown in Fig. 5 (e), crack skeletonization is to obtain the central axis of the image, and the width of the crack image is changed to unit pixel value through limited open operation and corrosion operation using morphological method. The equation is:

$$A \circ B = (A \ominus B) \oplus B \tag{12}$$

where A $\ominus$ B represents an erosion operation on image A with convolution kernel B; $\oplus$ represents an expansion operation.

B. Morphological Calculation

During pavement image processing, the accuracy of edge placement is affected by various noises in the image due to the difference in pavement material texture and imaging conditions. If the edges are not positioned accurately, then the pixel-by-pixel calculation of the crack width will be incorrect. In addition, pavement texture and noise disturbances may be different around adjacent crack pixels, which may lead to spurious differences in crack widths computed in the pixel direction. Therefore, the crack width obtained on the crack curve usually shows large fluctuations, and these widths do not necessarily reflect the actual situation of the crack, as show in Fig. 6.

This refinement technique allows the crack skeleton boundary output from the image contour detection network to become more refined to a single-pixel-width skeleton, as show in Fig. 7. By traversing each skeleton line at the pixel level, the width diameter w_i of each crack can be calculated,

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which helps to analyze and measure the characteristics of the crack more precisely, and the length of the crack can be calculated by Eq. (13).

$$l = \sqrt{\frac{\frac{1}{m}\sum_{i=1}^{m}(w_i - \bar{w})^2}{\bar{w}}} \times 100\% \tag{13}$$

where l is the longest distance (in pixels) in the horizontal direction in a single crack image, m is the row size of the pixels in the crack image, w_i is the diameter of the i-th crack, and $\bar{w}$ represents the average of the diameters of all cracks in the crack image. Therefore, the total variable coefficient L of the crack length can be calculated by Eq. (14).

$$L = \frac{\sqrt{\frac{1}{R}\sum_{j=1}^{R}(\bar{w}_j - \bar{D})^2}}{\bar{D}} \times 100\% \tag{14}$$

where R represents the total number of images captured in the time series, $\bar{w}_j$ is the average diameter of the crack segment in the i-th image, the diameter of the total crack is denoted as $\bar{D}$.

During contour inspection, different methods are used to generate different thicknesses of boundary lines. After the refinement process, the detected diameter may produce some scaling error.

$$w_i^* = \alpha w_i \tag{15}$$

where α is the scaling ratio, w_i is the actual diameter of the i-th crack, w_i^* represents the visually detected diameter.

Then, the average width diameter $\bar{w}^*$ of visual inspection can be calculated by

$$\bar{w}^* = \frac{\sum_{i=1}^{m}\alpha w_i}{m} = \alpha\bar{w} \tag{16}$$

Therefore, the uniformity of the visually detected crack length $cv(l)^*$ can be calculated according to Eq. (17).

$$cv(l)^* = \frac{\sqrt{\frac{1}{m}\sum_{i=1}^{m}(\alpha w_i - \alpha\bar{w})^2}}{\alpha\bar{w}} \times 100\% = cv(l) \tag{17}$$

The uniformity of visual inspection is achieved by calculation at the pixel level, so it will not be affected by the scaling ratio of the image. When calculating uniformity, the thickness or position of the boundary line will not affect the accuracy of the uniformity test results.

C. Area Measurement

In block cracks usually contain multiple fragments, each with a corresponding block size property. The so-called block refers to the longest distance from one point to another point within a single closed contour area of network fractures. The block sizes are indicated by lines in Fig. 8.

The following are the precise solution steps:

(1) Refinement of the cropped block crack image to obtain the skeleton of the cracks;

(2) Scan the refined alligator crack image line by line, count the total number of pixels in the i-th crack row $P(i)$, and record the coordinates of each pixel point in that crack row $(i, Z(i, k))$, $k = 1,2,\ldots, P(i)$, subtracting the vertical coordinates of adjacent pixel points in the crack interval of the i-th row two by two to obtain the corresponding coordinate differences: $L_i = (Z_{12}, Z_{23}, \ldots, Z_{k(k-1)})$;

(3) The refined block crack image is again scanned column by column, and the total number of pixels in the j-th column of cracks is counted as $P(j)$, while the coordinates of each pixel point in that column of cracks are recorded $(H(k, j), j)$, $k = 1,2,\ldots, P(j)$, and subtract the horizontal coordinates of adjacent pixel points in the j-th crack interval by two to obtain the corresponding coordinate difference: $W_j = H_{12}, H_{23}, \ldots, H_{(k-1)k}$;

(4) Finally, the actual length of the maximum of L_i and W_j is taken as the blockiness K of the reticulation crack, i.e:

$$K = [\max\left(\max\left(L_i\right), \max\left(W_j\right)\right)] \times \eta \tag{18}$$

where η is the actual length corresponding to the edge length of each pixel.

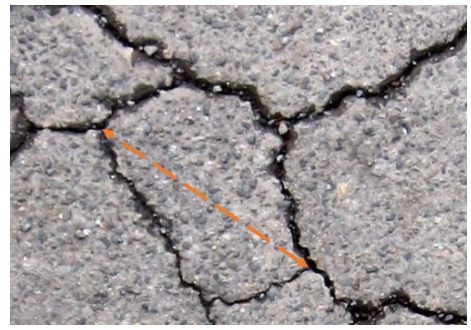

Fig. 8. Block diagram.

V. Experiment and Analysis

A. Comparison of SSA-CNN Model Optimization

When using the SSA to optimize the CNN, it is first necessary to select the hyperparameters to be optimized, such as the number of iterations of the training set, the learning rate and the discard rate. Then, according to the selection criteria within the predetermined hyperparameters, random combinations of hyperparameters are created, and each combination of hyperparameters represents a neural network. The accuracy of the validation set data is used to calculate the fitness function, where the classification accuracy of the crack type network is more important. The hyperparameter combinations of individuals are placed into the network for training to obtain the fitness of each individual. The population size of the SSA is 40, where the percentage of discoverers is 0.4 and the percentage of scouts is 0.2. When the fitness value reaches the preset warning value of 0.8, the iteration is stopped and the individual with the highest fitness is output, which is the optimal hyperparameter combination. The whole optimization process of the SSA iterates 500 times, and if the preset warning value is still not reached when the maximum population number is reached, the next round of iterations is performed. In this way, the hyperparameters of the neural network can be effectively optimized, thus improving its performance and accuracy.

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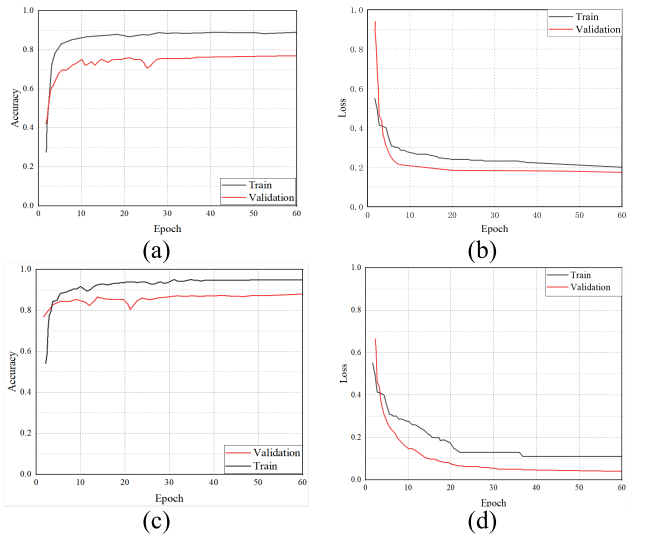

Fig. 9. a) CNN model accuracy; b) CNN model loss; c) SSA-CNN model accuracy; d) SSA-CNN model loss.

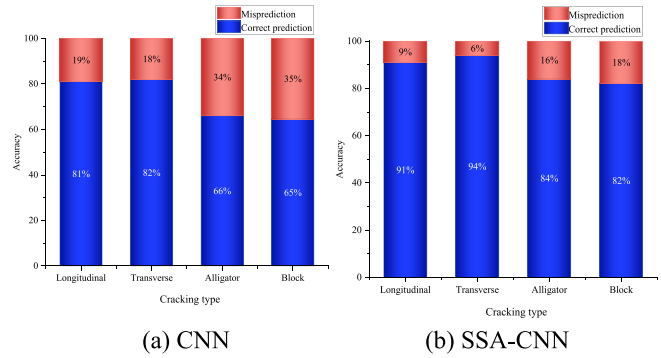

Fig. 10. Accuracy of crack image recognition.

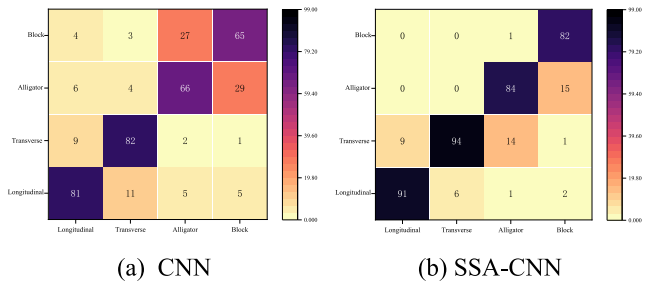

Fig. 11. Confusion matrix diagram.

In the process of training neural network model, the validation set is often used to evaluate the performance of the model. From Fig 9(a), it can be found that the accuracy rate of the validation set decreases in the early stage of training, then reaches the lowest point in 25 training epochs, and then gradually increases. The accuracy rate of the training set shows an overall increasing trend and tends to be stable after 25 training epochs, which indicates that the model has fully learned the training data. At the same time, in Fig. 9(b), the loss of the training set has a general downward tendency, particularly in the initial training epochs, and it tends to stabilize after 25 training epochs. Combined with the accuracy rate chart of the validation set, it can be found that the accuracy of the validation set is generally on the rise, and gradually stabilizes at around 90% after 25 training epochs. However, it is important to note that after 25 epochs, the loss on the validation set starts to increase due to over-training causing the model to overfit the validation set data. Therefore, manual tuning of hyperparameters is not easy to keep the model optimal.

The optimized model trained on the pavement crack dataset exhibits better training and validation loss function curves. In Fig. 9(c), the accuracy on the training set increases significantly in the first 5 epochs, and then gradually stabilizes, reaching 98.23% after 15 epochs. The accuracy on the validation set fluctuates during the first 15 epochs, after which it stabilizes at around 0.896. It is shown in the training set of Fig. 9(d) that the loss function drops rapidly and stabilizes at 0 after 15 epochs. While the initial loss of the validation set is 0.183, the overall decline is slow, and it stabilizes at around 0.07 after 15 epochs. Compared with the pre-optimized model, the optimized model only needs to be trained for 15 epochs to reach a relatively stable state, thus saving 10 training epochs.

The superimposed bar graph can clearly show the accuracy of each crack image recognition. In the test set of Fig. 10(a), the identification accuracy rates of transverse cracks and longitudinal cracks are 81% and 82%, respectively, while the accuracies of block cracks and alligator cracks are slightly lower, 66% and 65%, respectively. Combined with the confusion matrix below, it can be analyzed that some network cracks and block cracks with similar shapes are easily confused, so the accuracy of these two types is relatively low. The superimposed bar graph of the SSA-CNN model is shown in Fig. 10(b), and the recognition accuracy rates of longitudinal cracks, transverse cracks, block cracks and alligator cracks are 91%, 94%, 84% and 82%, respectively. Compared with the CNN model, the accuracy of the SSA-CNN model increased by 10%, 12%, 18% and 17%. In other words, the SSA-CNN model performed better in crack image recognition. Overall, the average accuracy of CNN and SSA-CNN crack image recognition correctness is 73.5% and 87.75%, respectively, an improvement of 14.25%.

The confusion matrix is a prominent technique for assessing network model performance in deep learning tasks like classification and recognition. The higher the value on the diagonal in the confusion matrix, the more samples the model correctly classifies, which means the better the performance of the model. In Fig. 11(a), the comparison of actual predicted values and SSA-CNN model predicted values for different types of fractures is shown. The figure shows that in the classification of alligator cracks and block cracks, the recognition accuracy is relatively low, and the proportion of correct predictions is about 66% and 65%. Meanwhile, 29% of block cracks were mispredicted as alligator cracks, and 27% of alligator cracks were misjudged as block cracks, which are consistent with the results in Fig. 10(a). The confusion matrix of the SSA-CNN model is shown in Fig. 11(b), and it demonstrates that the SSA-CNN model significantly improves the recognition accuracy of block cracks and alligator cracks. At the same time,

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TABLE II
COMPARISON OF PERFORMANCE BETWEEN DIFFERENT MODELS

Comparison	Recall	F1	Precision	mIoU
Mei et al. [37]	70.6%	75.4%	84.9%	-
Yu et al. [38]	86%	80%	75%	-
Kang et.al [26]	92.7%	92.2%	92.7%	92.6%
Shi et.al [39]	77.2%	81.45%	89.51%	69.19%
YOLO[5]	87.1%	87.8%	88.51%	-
Our method	87.1%	84.2%	96.2%	90.6%

no other confusion occurs in the predicted and actual values for transverse cracking and longitudinal cracking. However, due to the similar morphological characteristics of block cracks and network cracks, there is a certain degree of subjectivity and uncertainty in the classification of training and test sets. This caused some fissures and block cracks to be classified confounded, so the classification accuracy of alligator cracks and block cracks was lower than that of linear cracks.

Different crack classification algorithms usually use three metrics to evaluate the classification model: precision, recall and F1 score, as shown in Eqs. (19)-(21). Among them, the precision indicates how many of the objects detected by the model are real objects, and the recall rate indicates how many of all the real objects detected by the model are correctly detected. However, the accuracy of this crack assessment is not appropriate because the size of the cracks is usually too small compared to the background scene; therefore, higher scores are usually provided if the crack size is small. Precision and recall do not account well for false-positive and false-negative detections, and F1 scores can control these through parameter changes. The mean intersection ratio (mIoU) is one of the more reasonable and accurate evaluation metrics, which can more accurately consider false positives and false negatives. Therefore, many studies in the fields of computer vision and deep learning have also used mIoU as an evaluation metric for their models. Table II compares the proposed model with other methods mentioned in the literature, it can be seen that the precision, recall and F1 score of SSA-CNN are almost higher than other methods, and the performance has also been improved to a certain extent. The generalization ability of the model is strong, especially the accuracy is significantly higher than other models. Although YOLO has a slight advantage in F1 values, our algorithm performs significantly better than YOLO in Precision, which indicates that our algorithm has more advantages in the accuracy of prediction results and positioning accuracy. In addition, future work can improve the performance of crack detection by improving hardware performance and model optimization.

$$Recall = \frac{TP}{TP + FN} \tag{19}$$

$$Precision \frac{TP}{TP + FP} \tag{20}$$

TABLE III
COMPLEXITY OF DIFFERENT METHODS

Models	Params(M)	Training Times(S)	Average inference time (MS)
CNN	42.36	17424.1	24.62
DeepLabv3	39.63	16244.4	22.64
FCN	32.95	12991.8	20.37
ECSNet[40]	0.41	5011.2	13.66
SSA-CNN	**28.42**	**8429.61**	**18.4**

$$F1score = \frac{2 \times precision \times recall}{precsion + recall} \tag{21}$$

$$mIoU = mean(\frac{TP}{TP + FP + FN}) \tag{22}$$

where TP is the true positive, FP is the false positives, and FN is false negative.

Table III presents the parameters of each model. SSA-CNN maintains good performance while having a lower number of parameters, which means it is more lightweight. Compared to traditional CNNs and DeepLabv3, SSA-CNN has a lower training time, which means that SSA-CNN is easier to train in resource limited situations. Although the inference time is slightly higher than ECSNet, SSA-CNN still has a faster inference speed compared to traditional CNN and DeepLabv3, making it more suitable for real-time task applications. Overall, SSA-CNN has made significant progress in efficiency, although there is still some room for improvement, it has demonstrated significant advantages compared to traditional models.

B. Comparative Analysis of Compensation Results

Select a tunnel with typical cracks for comparative testing, the tunnel length is about 550m, and a total of 1200 frames of images are collected, with each image size of 256 × 256. The acquired images are processed using the improved multi-scale Retinex algorithm, and some of the original images with typical cracks are compared with the images processed by this algorithm as shown in Fig. 12. It can be seen that the proposed image compensation method for tunnel pavement detection can improve image quality, reduce the impact of collection platform vibration on the image, and provide a good foundation for subsequent quantitative detection of tunnel diseases and cracks.

In order to accurately evaluate the performance of the compensation method, the peak signal to noise ratio (PSNR) index is chosen to quantitatively evaluate the effect of the image before and after compensation, and the formula for calculating PSNR is:

$$P_s\left(I_a, I_b\right) = 10lg\frac{255^2}{MSE\left(I_a, I_b\right)} \tag{23}$$

where $MSE\left(I_a, I_b\right)$ is the mean square difference between the reference frame I_a and the target frame I_b in the image sequence. When the quality of the compensated image is good, the PSNR is larger than the initial value, and the larger the difference is, the better the compensation effect is.

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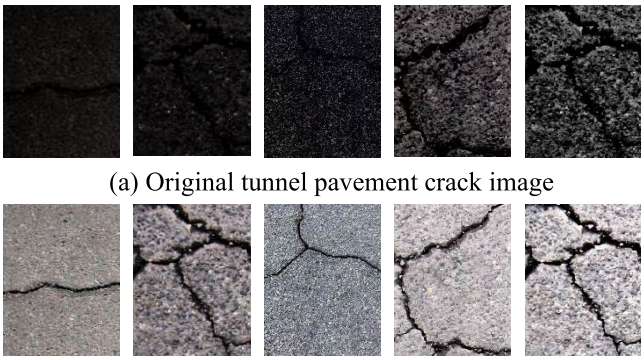

(a) Original tunnel pavement crack image

(b) After optimization of multi-scale Retinex algorithm

Fig. 12. Image comparison before and after optimization.

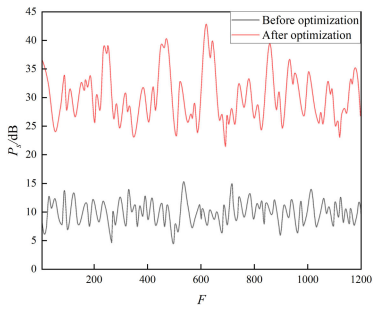

Fig. 13. PSNR results of the compensation method.

When the PSNR value is greater than 30 dB, the image quality after denoising can be considered good, and less than 20 dB indicates that the image quality is unacceptable. The PSNR of 1200 frames is calculated, and the results are shown in Fig. 13. The PSNR of the images after the compensation process using this method are higher than those before the compensation, and the average PSNR is 34.61 dB, and the maximum is 43.24 dB, which indicates that the image quality is improved. This method has good correction performance for tunnel pavement images and can compensate for the errors in image acquisition.

C. Classification of Crack Damage Levels

In the current technical specification (JTG5210-2018), the classification of cracks in China is mainly based on the morphology of cracks, which is divided into four categories: alligator cracks, block cracks, longitudinal cracks and transverse cracks. According to the size of crack blockiness or width it is divided into three grades of light, medium and heavy, and finally subdivided into 9 subcategories, as shown in Table IV.

After completing the classification of cracks and calculating the length and width of cracks, it is possible to achieve the evaluation of the degree of damage to cracks in the current standard, as shown in Table V.

The maximum width of linear cracks is extracted after binarization, denoising, edge detection, and skeletonization, and the damage level of the cracks is rated. Finally, by including an external rectangle, the crack length is determined, and the detection error is computed. To extract the block size of the alligator crack and determine the error, the alligator crack is binarized, threshold denoised, and then closed with the

TABLE IV
EVALUATION CRITERIA FOR THE DEGREE OF CRACK DAMAGE

Type (i)	Damage name	Classification Criteria	Grade
1	Alligator Cracking	The block size is between 0.2-0.5m, and the crack width is ≤ 2mm	Low
2		Block size ＜ 0.2m, crack width between 2-5mm	Medium
3		Block size ＜ 0.2m, crack width ＞ 5mm	High
4	Block Cracking	Block size ＞ 1.0m, crack width between 1-2mm	Low
5		The block size is between 0.5-1.0m, and the crack width is ≥ 2mm	High
6	Transverse Cracking	Crack width ≤ 3mm	Low
7		Crack width ＞ 3mm	High
8	Longitudinal Cracking	Crack width ≤ 3mm	Low
9		Crack width ＞ 3mm	High

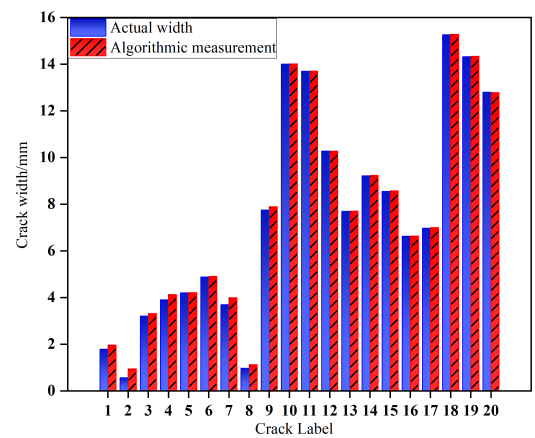

Fig. 14. Bar chart comparing the error of crack average width measurement.

addition of an exterior rectangle. Then, take out the skeleton of the fracture, figure out its average width, and assign it a damage level based on the size of the block. The results of the form parameter and image processing for linear cracks are displayed in Table VI.

By comparing the maximum crack width with the actual measured value, the error is computed. Table V shows that the average accuracy rate is 92.07%, the average width error is 7.93%, the average minimum error is 3.47%, and the highest error is 12.6%. Table V shows the morphological parameters and calculation results for processing transverse crack and longitudinal crack images.

In this paper, the experiment of crack width measurement is carried out, and multiple cracks are measured using the skeleton boundary refinement algorithm. Fig. 14 shows the experimental measurements of 20 representative cracks and plots the bar graphs against the manual measurements of crack widths and the algorithmic results. The study discovered that

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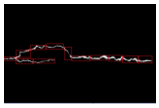
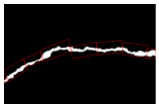
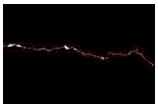
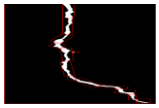
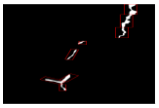
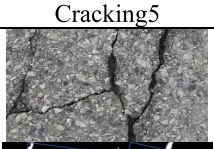
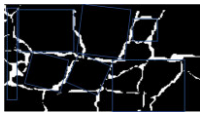
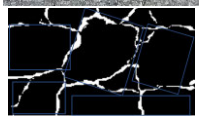
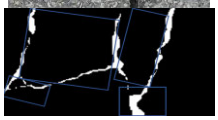
TABLE V
CALCULATED RESULTS OF LINEAR CRACK PROCESSING IMAGES AND MORPHOLOGICAL PARAMETERS

Images						
Type	T	T	T	L	L	L
Bounding Rectangle						
Width (mm)	1.210	9.682	2.672	8.756	1.953	12.654
Error (%)	3.47	12.6	10.42	5.63	5.78	9.7
Length(m)	0.402	0.671	0.921	1.208	0.845	1.243
Grade	Low	High	Low	High	Low	High
mIoU (%)	89.4	90.2	91.7	90.1	92.4	89.6

Note: T indicates transverse cracks, L indicates longitudinal cracks.

TABLE VI
CALCULATED RESULTS OF THE IMAGES AT THE ALLIGATOR CRACKS

NO.	Cracking1	Cracking2	Cracking3	Cracking4	Cracking5
Original					
Bounding Rectangle					
Type	Block	Alligator	Alligator	Alligator	Block
Grade	Moderate	High	High	High	Moderate
Area (m^2)	0.381	0.892	0.756	0.847	0.342
Block (m)	0.395	0.57	0.691	0.604	0.283
Width (mm)	1.823	2.511	3.724	2.761	2.859
Error(%)	7.2	2.4	8.4	12.9	13.1
mIoU(%)	85.9	88.3	87.6	90.1	89.7

the measuring error is significant when the crack width is less than 3 mm, and it is very tiny when the crack width is higher than 3 mm, with a relative error of less than 0.02. In particular, for broader cracks, the measuring error is almost negligible. In addition, the crack width measured using the skeleton boundary refinement algorithm has some continuity, while the manual test results have some dispersion. Therefore, measuring the fracture width by the skeleton boundary refinement algorithm can detect the average width of the fracture more accurately. However, the measurement of the crack width is obtained by manual measurement with a crack width gauge, and the accuracy of the results may be affected by the accuracy of the instrument and the reading error of the human eye. In general, the skeleton boundary refinement algorithm is a more reliable fracture width measurement method, which can improve the accuracy and continuity of the measurement results.

In order to make a fair comparison therefore we have used two other methods with the proposed method in this paper, with the dataset of this paper, and the updated results are shown in Table VII. We compared the accuracy of different crack feature extraction methods with other models. The method has a smaller error margin and higher stability, which makes it ideal for engineering applications. However, the average accuracy of the method needs to be further improved compared to some models.

TABLE VII
COMPARISON OF CRACK FEATURE EXTRACTION ACCURACY

Comparison	Width (error)
A.K. Ji [41]	−19.68%~49.12%
X.C. Yang [8]	−16.21% ~ 29.4%
Our method	−11.72%~22.36%

VI. CONCLUSION

Automatic detection and segmentation of road cracks is a necessary prerequisite for road maintenance. Therefore, this

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article proposes a two-step road crack detection based on CNN to improve the performance of road crack detection. The main novelty of this method lies in the combination of improved SSA-CNN and skeleton refinement-based road crack detection methods. In the two-step method proposed in this article, the results of the first step of road crack detection can be used for the second step, so that the detected cracks can be purposefully calculated in the second step. Unlike mainstream YOLO series algorithms [42], most of them focus on crack detection but often overlook the spatial layout and positional details of crack images, making it difficult to convert the identified crack information into quantitative evaluation and connect it with actual maintenance work.

This method improves the accuracy of pavement crack detection and segmentation. The first step of detecting the classification and location of pavement cracks is used to improve the accuracy of the second step of segmenting the crack pixels from the complex pavement background. The accuracy of the second-step segmentation of detected cracks is affected by the performance of the first-step crack detection, which implies that the proposed two-step segmentation method can be further improved by developing crack detection and segmentation methods simultaneously.

Through the optimization of CNN hyperparameters by SSA, the improved method in this paper significantly improves the accuracy of crack detection, reaching 96.2% Precision, and provides a more comprehensive performance evaluation by introducing mIoU index. By comparing the number of parameters, inference time and model operation efficiency, the model is reduced by 33%, 52% and 25% respectively compared with traditional CNN.Aiming at the crack image quality problem under low illumination, an improved multi-scale Retinex network enhancement method is proposed, which effectively improves the PSNR of the image, with an average value of 34.61 dB and a maximum value of 43.24 dB, significantly improving the visual effect of the tunnel pavement image. By eliminating the influence of edge noise on the pixelated crack width measurement, the algorithm achieves stable crack width measurement. Compared with the existing researches, the error is only -11.72% to 22.36%.

Cameras capable of capturing fine features at high speeds may be required due to different datasets and different focal lengths of cameras, however, this is a common challenge for any deep learning model used to segment objects [43]. In addition, it is difficult to accurately identify whether it is a transverse or longitudinal crack in oblique cracks in engineering. The criteria for classifying transverse and longitudinal cracks for future work are also worth exploring. There are also improvements that can be made to the CNN, e.g., reducing the number of parameters that need to be learned in the model by parameter sharing to make it more lightweight in order to better demonstrate the advantages of combining SSA methods.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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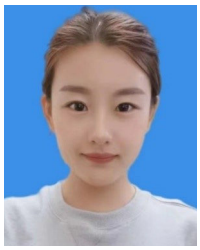

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